

NAME: \_\_\_\_\_

PERIOD: \_\_\_\_\_

## One Table, Two Study Designs

AP Statistics · Unit 3 · Topic 3.14

### [STAR] THE PROBLEM

Suppose a transit authority obtains the shown counts under either design.

**Design A:** Take independent SRSs of 100 from each of three route populations of 5,000 and record ticket method.

**Design B:** Take one SRS of 300 from 15,000 system riders and record both route and ticket method.

**Ari:** “Route rows imply homogeneity; the counts are enough to choose.”

**Bea:** “Read how riders were sampled; identical counts can support different population claims.”

**Ticket method**

| Route | Mobile | Card | Cash | Total |
|-------|--------|------|------|-------|
| North | 60     | 25   | 15   | 100   |
| East  | 48     | 32   | 20   | 100   |
| West  | 42     | 33   | 25   | 100   |
| Total | 150    | 90   | 60   | 300   |

### FIRST TAKE — one sentence before you work the parts

Do the same counts mean that riders were selected in the same way? Explain in one sentence.

### [BOOK] LET DATA COLLECTION NAME THE TEST (keep on the page)

Use chi-square homogeneity to compare one categorical distribution across independently sampled populations or randomized treatments:  $H_0$  says the distributions are the same, and  $H_a$  says at least one differs. Use chi-square independence for one random sample with two categorical variables:  $H_0$  says there is no association, and  $H_a$  says there is an association. Both use  $E = (\text{row total})(\text{column total})/(\text{table total})$ . Check the random design, each 10 percent condition when sampling without replacement, and every expected count  $> 5$ .

Work (a)–(c).

**DOK 1** (a) How many random samples are taken in Design A and in Design B?

**DOK 2** (b) Calculate the three expected counts for the North row. Explain briefly why the other two rows have the same expected counts.

**DOK 3** (c) In 4 sentences, judge Ari’s rule. Name and justify the test for each design, state each null hypothesis in context, and explain why the same table can lead to different test names.

**Frame:** Design A uses \_\_\_\_\_ because \_\_\_\_\_; its null says \_\_\_\_\_.

**Frame:** Design B uses \_\_\_\_\_ because \_\_\_\_\_; its null says \_\_\_\_\_.

Turn this sheet in whenever you finish — bonus credit. Part (c) is scored **E / P / I**.